

DEWBERRY AGCM, AB, Canada

WMO: 713590

Lat: 53.6592N

Lon: 110.5875W

Elev: 616

StdP: 94.14

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-31.9	-28.9	-35.5	0.1	-31.7	-32.2	0.2	-28.8	8.2	-4.6	7.0	-7.8	1.4	280	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	27.7	17.4	26.1	16.8	24.6	16.3	19.6	25.1	18.5	23.9	17.4	22.6	2.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	13.6	22.5	16.4	12.5	21.2	15.4	11.7	19.9	58.6	25.1	54.9	24.2	51.4	22.6	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	6.4	5.6	DB	-37.2	30.7	2.0	1.3	-38.6	31.6	-39.8	32.4	-40.9	33.1	-42.4	34.1	(4)
(5)			WB	-36.9	22.2	2.1	1.5	-38.5	23.3	-39.7	24.1	-41.0	25.0	-42.5	26.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-12.3	-13.1	-5.7	2.9	11.0	15.1	17.1	16.0	11.3	3.5	-5.9	-12.8	(6)
(7)		DBStd	12.77	8.55	8.34	7.47	5.35	4.53	2.97	2.40	2.89	4.66	4.81	7.02	7.89	(7)
(8)		HDD10.0	3522	692	647	486	219	44	3	0	1	40	207	478	706	(8)
(9)		HDD18.3	5872	950	880	744	464	229	104	52	82	214	460	727	964	(9)
(10)		CDD10.0	724	0	0	0	5	74	155	221	187	78	4	0	0	(10)
(11)		CDD18.3	34	0	0	0	0	2	6	15	9	2	0	0	0	(11)
(12)		CDH23.3	623	0	0	0	3	83	92	194	158	92	0	0	0	(12)
(13)		CDH26.7	98	0	0	0	0	12	12	28	23	24	0	0	0	(13)
(14)	Wind	WSAvg	2.4	2.5	2.4	2.6	2.9	2.7	2.4	2.1	1.9	2.1	2.5	2.3	2.2	(14)
(15)	Precipitation	PrecAvg	416	18	13	16	29	43	73	79	58	35	18	16	19	(15)
(16)		PrecMax	605	42	27	44	91	101	132	166	148	97	38	54	33	(16)
(17)		PrecMin	223	5	4	4	2	6	15	23	9	2	2	2	7	(17)
(18)		PrecStd	88	10	6	9	21	26	30	38	25	24	10	12	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	5.2	12.2	22.6	28.2	28.3	29.4	29.2	29.7	20.9	12.5	5.2	(19)
(20)			MCWB	3.0	2.3	5.9	10.2	13.8	16.7	20.5	18.1	16.6	12.3	6.2	1.8	(20)
(21)		2%	DB	3.9	2.9	8.3	18.3	25.8	25.9	27.1	26.8	26.1	16.2	7.9	2.8	(21)
(22)			MCWB	1.7	0.7	3.6	8.1	12.7	15.9	19.5	18.1	15.4	8.9	3.9	0.4	(22)
(23)		5%	DB	2.3	1.0	5.4	15.1	23.1	24.0	25.5	25.1	22.9	13.5	4.6	0.6	(23)
(24)			MCWB	0.5	-0.6	2.4	6.9	11.9	15.1	18.3	17.1	14.1	7.8	2.1	-1.3	(24)
(25)		10%	DB	0.2	-0.6	3.4	11.9	20.7	22.2	23.9	23.4	19.9	11.0	2.6	-2.0	(25)
(26)			MCWB	-1.3	-1.9	1.1	5.2	10.9	14.3	17.3	16.6	12.7	6.8	0.9	-3.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	2.6	6.7	10.5	15.2	18.6	22.1	21.1	17.4	12.2	7.3	2.2	(27)
(28)			MCDWB	5.7	4.8	11.5	22.2	24.0	25.1	28.1	26.5	26.9	19.4	11.6	4.7	(28)
(29)		2%	WB	1.9	0.9	4.0	8.6	13.7	17.3	20.2	19.3	16.0	10.1	4.4	0.5	(29)
(30)			MCDWB	3.7	2.4	7.5	17.1	22.7	23.3	25.7	24.7	24.6	13.8	7.5	2.7	(30)
(31)		5%	WB	0.5	-0.3	2.5	7.1	12.7	16.4	19.0	18.1	14.7	8.6	2.5	-1.3	(31)
(32)			MCDWB	2.2	1.0	5.2	14.4	21.1	21.8	24.3	23.6	21.8	12.8	4.3	0.3	(32)
(33)		10%	WB	-1.4	-2.1	1.3	5.7	11.7	15.3	18.0	17.0	13.2	6.9	0.9	-3.2	(33)
(34)			MCDWB	0.1	-0.7	3.3	11.1	19.5	20.3	22.9	22.2	19.0	11.0	2.3	-2.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.3	9.9	9.6	10.9	13.5	11.4	11.6	12.3	12.7	9.8	8.0	8.1	(35)
(36)			MCDBR	10.1	10.6	11.4	16.9	17.7	14.8	14.3	15.5	18.3	14.6	11.1	10.7	(36)
(37)			MCWBR	8.4	9.0	7.7	8.7	7.3	6.6	7.6	8.0	8.5	8.4	7.6	8.7	(37)
(38)		5% WB	MCDWR	9.9	10.3	10.9	16.2	15.6	12.4	13.8	13.9	17.5	13.7	10.6	10.3	(38)
(39)			MCWBR	8.3	8.8	7.4	8.6	6.8	6.3	7.8	7.5	8.6	8.4	7.6	8.6	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.243	0.291	0.282	0.328	0.348	0.367	0.363	0.373	0.310	0.275	0.250	0.216	(40)
(41)		taud		2.316	2.191	2.382	2.376	2.367	2.338	2.380	2.325	2.491	2.560	2.446	2.354	(41)
(42)		Ebn at Noon		754	800	913	904	899	881	878	845	872	843	753	735	(42)
(43)		Edh at Noon		55	88	91	106	113	118	111	111	82	61	49	43	(43)
(44)	All-Sky Solar Radiation	RadAvg		0.96	2.01	3.52	4.68	5.81	5.93	6.10	4.98	3.50	2.01	0.96	0.70	(44)
(45)		RadStd		0.10	0.15	0.28	0.40	0.42	0.43	0.33	0.37	0.38	0.29	0.11	0.07	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page